LEAPPs FIELD GUIDE

regfprobe User Manual

Reading Windows registry hives offline, choosing keys, and reporting on them: hives and disk images, search and timeline, HTML reports from one or many hives, and key profiles you can pull from any E01.

regfprobe 0.3.0
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Source and latest version: github.com/abrignoni/regfprobe

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1. What regfprobe is

regfprobe reads Windows registry hive files (SAM, SYSTEM, SOFTWARE, SECURITY, NTUSER.DAT, UsrClass.dat, Amcache.hve and others) **offline and read-only**. It never writes to a hive, its logs, or the image it reads them from. It can open a loose hive file, or reach straight into an E01 or raw disk image and read the hive from there, along with its `.LOG1`/`.LOG2` transaction logs.

It is written in plain Python with no packages to install. The same reading engine (`regfprobe.py`) runs the window, the command line, and every export, so the data you see on screen is the data that ends up in a report.

2. Getting started

What you need

- Python 3.10 or newer (from python.org on Windows; it includes Tkinter).
- These files together in one folder:

File	What it does
`regfprobe.py`	The hive reader and command line
`regfprobe_gui.py`	The window
`regfprobe_theme.py`	Dark mode
`regfprobe_report.py`, `regfprobe_interpret.py`	HTML reports and the Interpreted column
`regfprobe_profile.py`	Key profiles
`qnxprobe.py`, `ewfprobe.py`	Reading disk images (from the qnxprobe and ewfprobe projects). Only needed to open hives inside E01/raw images.
`docs/`	This manual and its screenshots

Starting it

```
python regfprobe_gui.py
python regfprobe_gui.py "C:\exports\SYSTEM"
python regfprobe_gui.py --image "D:\Cases\CASE-001.E01"
python regfprobe_gui.py "C:\exports\SOFTWARE" --key "Microsoft\Windows
NT\CurrentVersion"
```

With a hive or `--image`, the window opens straight to it; `--key` jumps to a key once the hive is open.

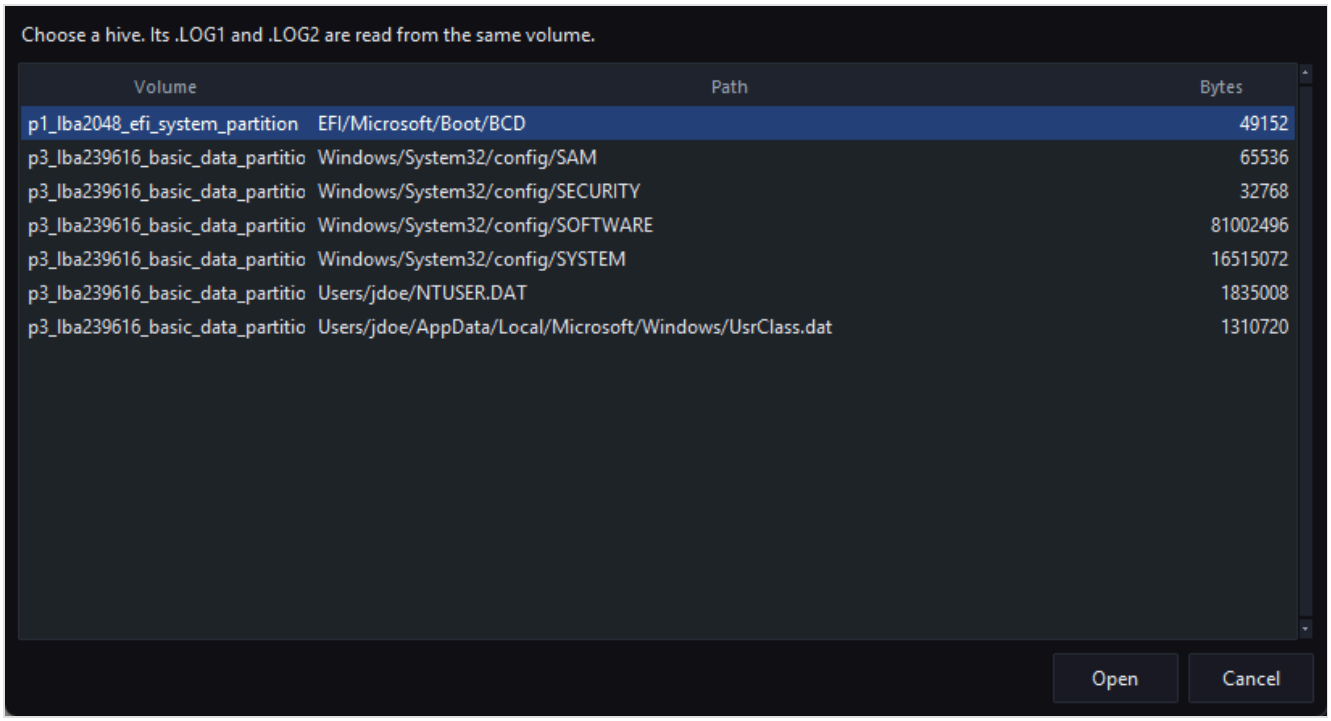
3. Opening hives

A hive file

File → Open Hive... (`Ctrl` + `O`) opens a hive you have already exported. Its `.LOG1` and `.LOG2` are picked up automatically when they sit beside it.

A hive inside a disk image

File → Open Hive from Disk Image... (`Ctrl` + `Shift` + `O`) asks for an E01 or raw image, searches every volume in it for files named like hives, and lists them. Choose one and press **Open** (or double-click it). The hive is read straight from the image into memory; nothing is extracted to disk.



The hive picker: volume, path inside the volume, and size of each hive found in the image.

Transaction logs

Windows writes registry changes to the `.LOG1`/`.LOG2` files first and folds them into the hive later, so a hive taken from a running or crashed machine is often "dirty". By default regfprobe replays the logs in memory, so you also see the changes they hold that were not yet in the hive. To see the hive exactly as stored on disk instead, untick **File → Replay Transaction Logs When Opening** before opening it.

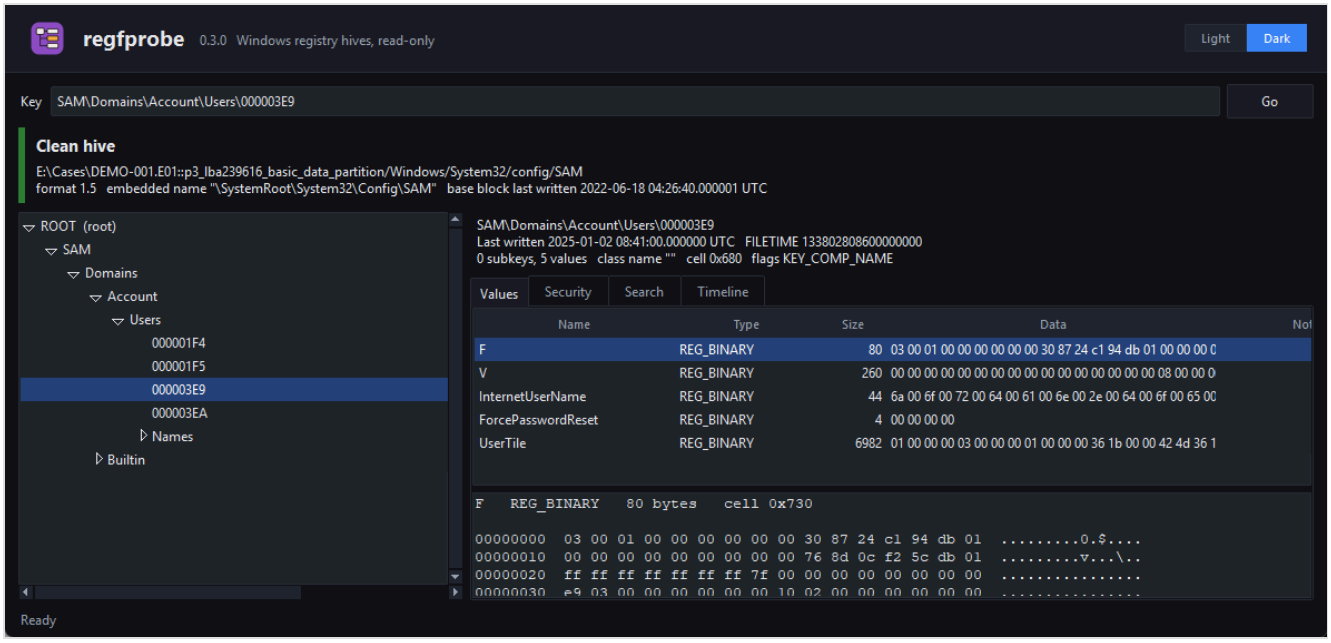
4. How far to trust a hive

The coloured banner under the address bar says what state the hive was in and what regfprobe did about it:

Banner	Meaning
Clean hive	Stored cleanly; read as it is.
Dirty hive: transaction logs replayed in memory	The logs held newer data and it was applied. The banner says how many log entries were replayed, their sequence numbers and which logs they came from.
Dirty hive read as it is: the transaction logs were not applied	The hive is behind its logs, but they were missing or could not be used (for example a log that fails its checks, or a replay that would set a key's time back). The banner gives the reason.
Read without replaying transaction logs	Replay was turned off, so the logs were not read. The hive may or may not be dirty; regfprobe did not check.
Damaged structures were skipped	Some cells could not be read. Everything readable is shown; the banner counts what was skipped so far, and the count grows as more keys are opened.

The same banner is copied into every HTML report, per hive.

5. The main window



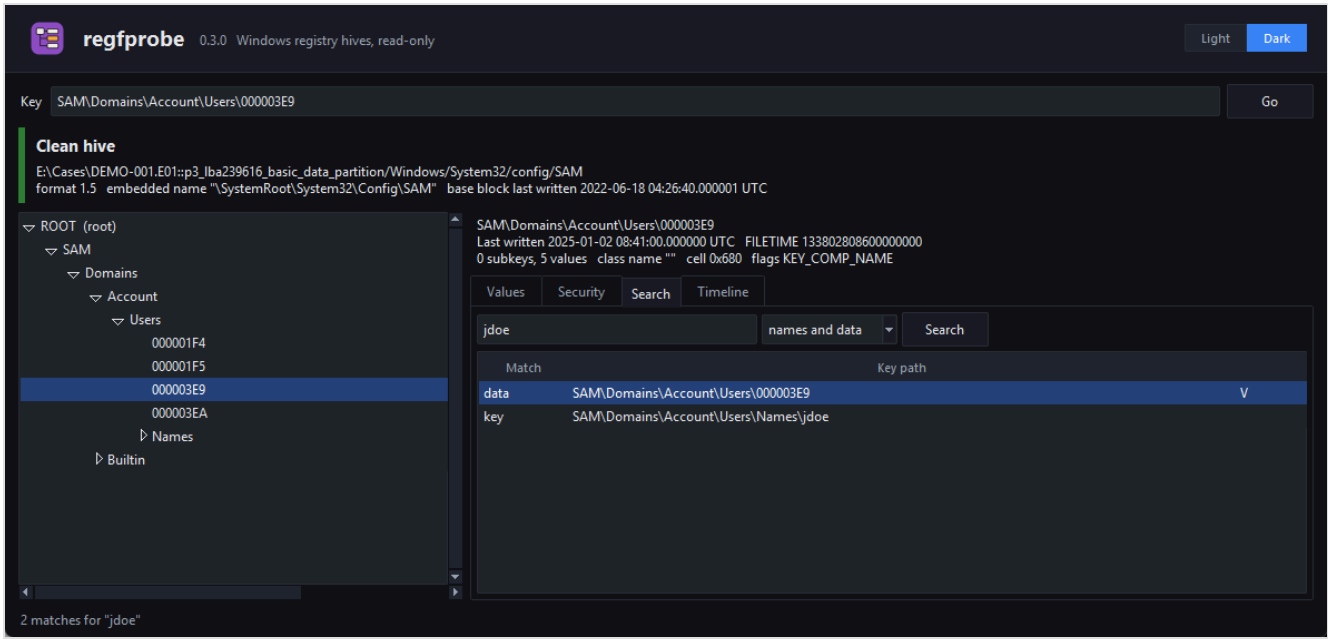
A SAM account key: the key tree on the left, the key's details and values on the right, the selected value's bytes below.

- **Key** (address bar): type or paste a key path and press `Enter` to jump to it (`Ctrl`+`L` focuses it).
- **Key tree**: expand with the arrows or `Enter`. Right-click a key to copy its path or name, export it, or add it to a report.
- **Key details**: last-written time (UTC and raw FILETIME), subkey/value counts, class name, cell offset and flags.
- **Values** tab: name, type, size, decoded data and notes. Click a column heading to sort. Select a value to see its decoded data and a hex dump in the lower pane.
- **Security** tab: the key's security descriptor (owner, group, permissions).
- **Search** and **Timeline** tabs: below.

The **Notes** column flags anything unusual about how a value is stored (for example a string with no terminator, or data shorter than its stated size).

6. Search

The **Search** tab (`Ctrl`+`F`) looks through the whole hive and stops at 5,000 matches (the status bar says so; narrow the search). Type the text, choose what to search (*names and data*, *names* only, or *data* only), and press **Search** or `Enter`. Double-click a result, or press `Enter`, to jump to that key.



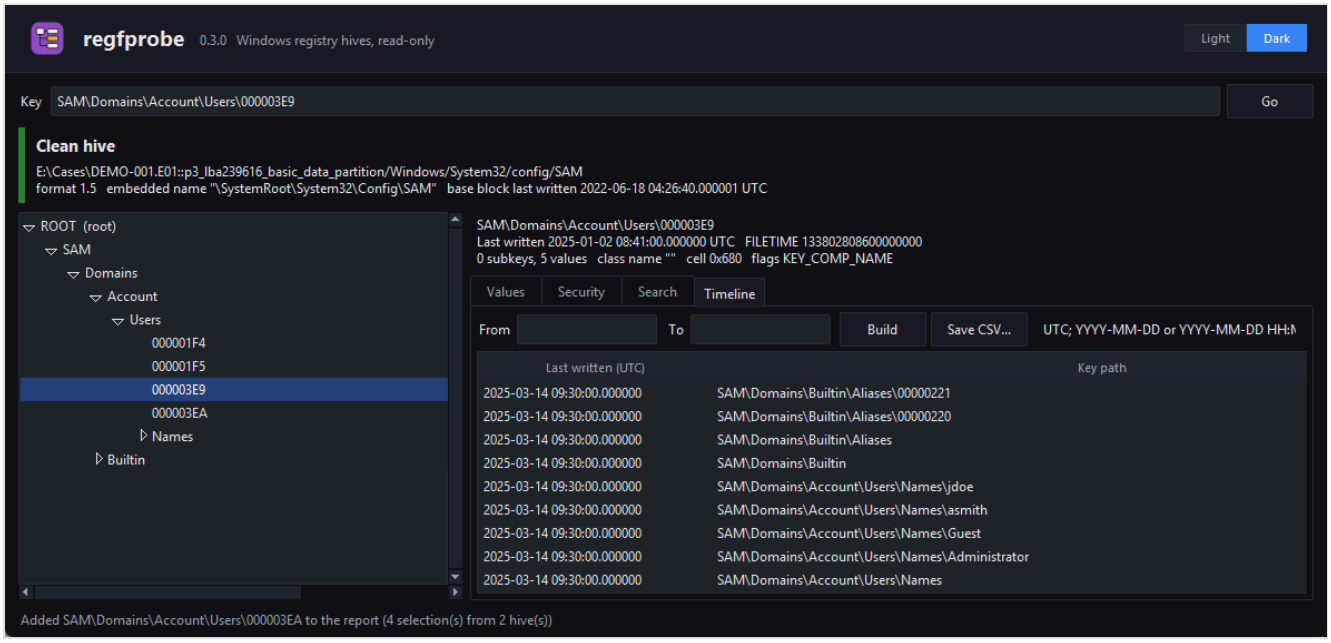
Searching for `jdoe`: one match in a key name and one inside a value's data.

Long tasks

Opening a large image, searching and building the timeline run in the background; the status bar shows progress. Press `Esc` to cancel.

7. Timeline

The **Timeline** tab lists every key by its last-written time, newest first. Fill in **From** and/or **To** (UTC, `YYYY-MM-DD` or `YYYY-MM-DD HH:MM:SS`) to narrow it, and press **Build**. Long timelines are shown a page at a time. Double-click a row to open that key. **File → Save Timeline as CSV...** writes the whole timeline to a spreadsheet-friendly file.



Every key by last-written time, newest first.

> A key's last-written time changes when one of its values is set, changed or deleted, and when a subkey is created or deleted under it. It does not tell you *which* change it was, and it does not change when a subkey's values change.

8. Copying and exporting

To get	Use
A key's path or name	**Edit → Copy Key Path** (`Ctrl`+`Shift`+`C`), or right-click the key
A value's data, name or bytes	**Edit → Copy Value Data** (`Ctrl`+`C`), Copy Value Name, Copy Value Bytes as Hex, or right-click the value
A key and everything under it as data	**File → Export Selected Key...** (`Ctrl`+`E`): JSON, CSV or a `.reg` file. Pick the format by the file extension.
Every key's last-written time	**File → Save Timeline as CSV...**
A readable report of chosen keys	The HTML report, next section

9. HTML reports

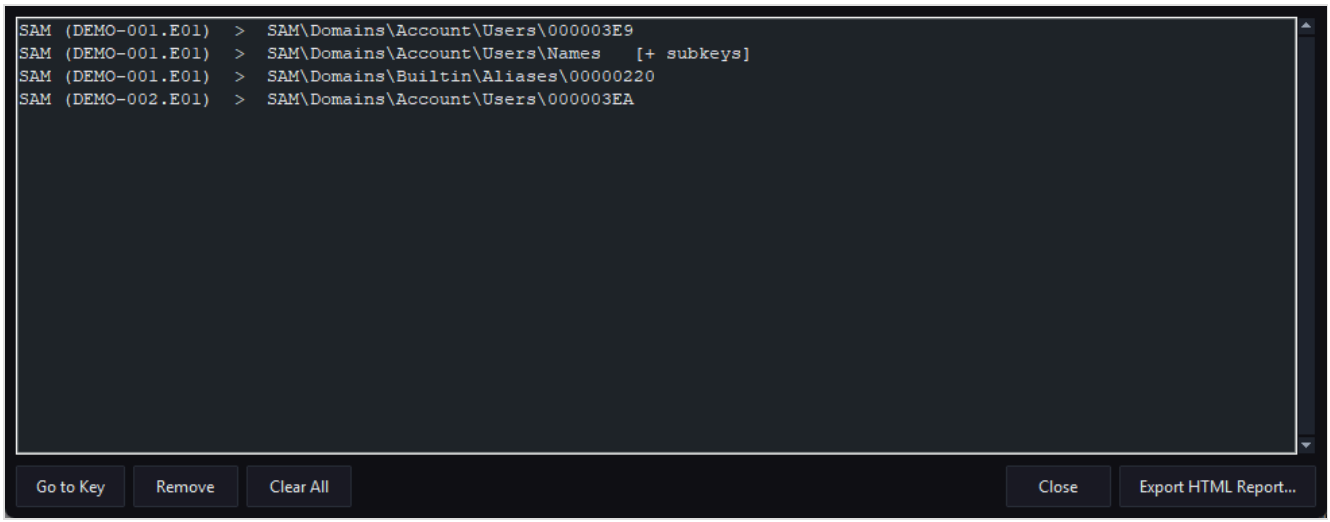
An HTML report collects the keys you choose, from one hive or many, into a single file that opens in any browser, prints cleanly, and needs nothing else to view: pictures and styles are inside the file.

Choosing keys

1. Select a key in the tree.
2. **Report → Add Selected Key** (`Ctrl` + `R`) adds just that key; **Report → Add Selected Key and Subkeys** adds it and everything under it. Both are also on the tree's right-click menu.
3. Open another hive, from the same image or a different one, and keep adding. The list is kept when you switch hives; the status bar counts selections and hives.

Reviewing the list

Report → Keys in Report... shows each selection with the hive and image it came from.

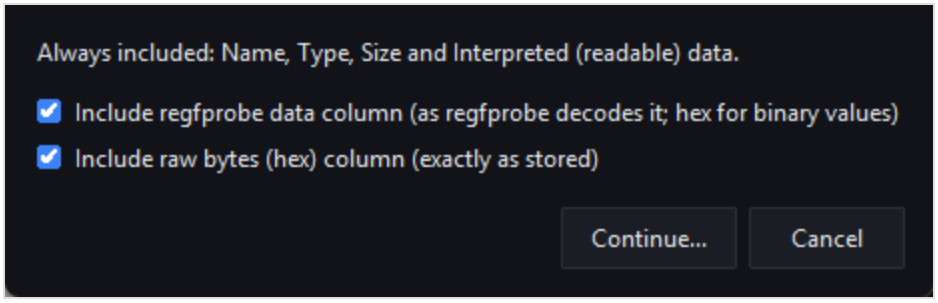


Selections from two images. `[+ subkeys]` marks a key added with everything under it.

- **Go to Key** (or double-click) switches to that hive and selects the key.
- **Remove** (or `Delete`) drops the selected lines; **Clear All** empties the list.
- **Export HTML Report...** goes straight to exporting.

Exporting

Report → Export HTML Report... (also on the File menu) first asks which optional columns to include, then where to save the file. Your choice is remembered for next time.

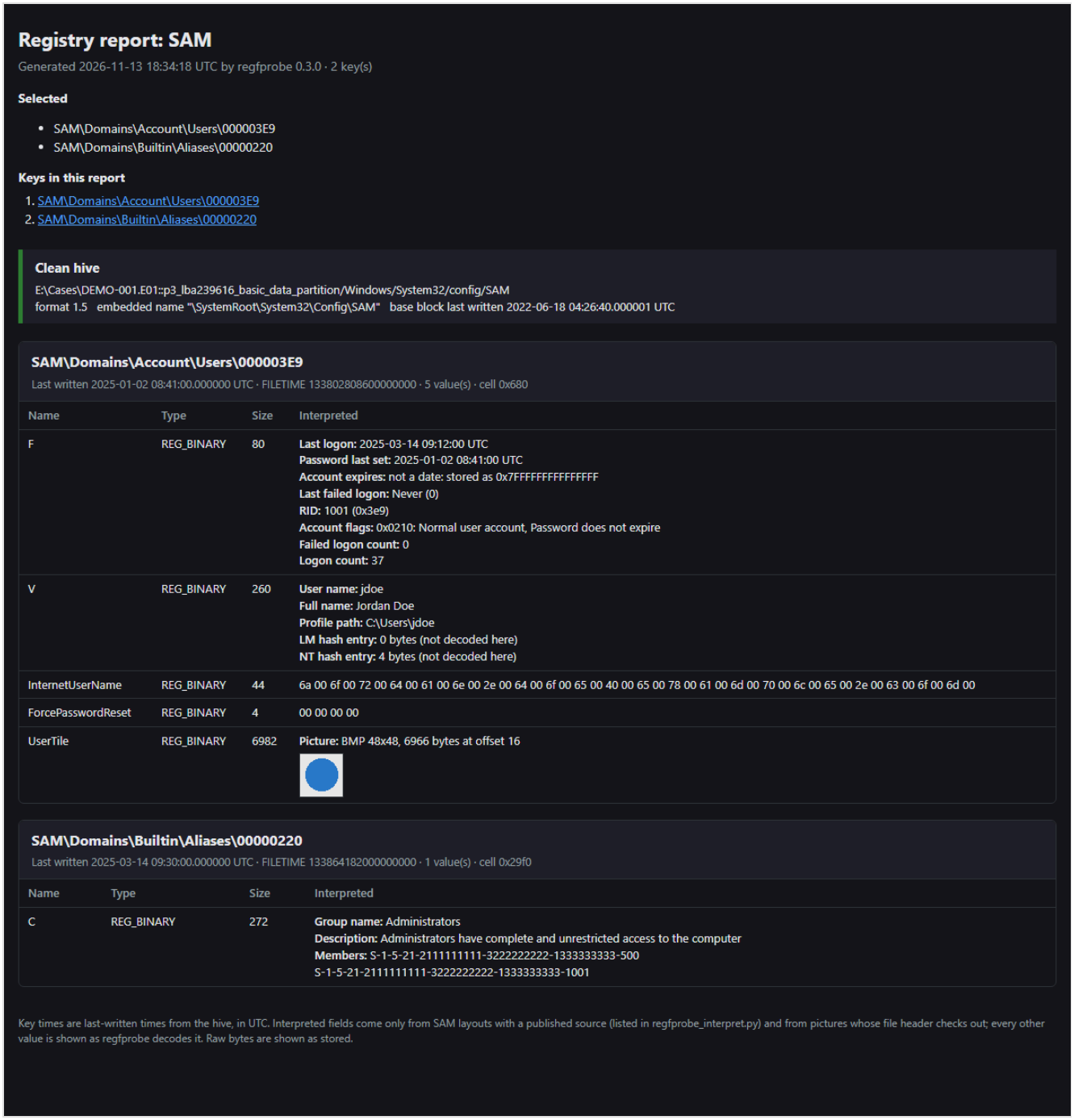


Name, Type, Size and Interpreted are always included.

Column	Shows	Include it when
Interpreted	The readable form: decoded SAM fields and pictures (next section). For strings and numbers, regfprobe's own decoding. Other binary values: their hex, or a dash when the regfprobe data column is included.	Always included
regfprobe data	The value as regfprobe's Values tab shows it. For binary values that is hex, so it repeats the raw column.	You want the report to match the window exactly
Raw bytes (hex)	The bytes exactly as stored (first 4 KB per value)	The report must let a reader check the decoding

What the report contains

- A title naming the hives, the time it was made, and the regfprobe version.
- An index: what you selected and links to every key, grouped by hive.
- Per hive: its full source (image, volume and path) and its trust banner.
- Per key: path, last-written time (UTC and FILETIME), value count and cell offset, then a table of its values.



Registry report: SAM

Generated 2026-11-13 18:34:18 UTC by regfprobe 0.3.0 · 2 key(s)

Selected

- SAM\Domains\Account\Users\000003E9
- SAM\Domains\Builtin\Aliases\00000220

Keys in this report

1. SAM\Domains\Account\Users\000003E9
2. SAM\Domains\Builtin\Aliases\00000220

Clean hive

E:\Cases\DEMO-001.E01::p3_lba239616_basic_data_partition/Windows/System32/config/SAM
format 1.5 embedded name "\SystemRoot\System32\Config\SAM" base block last written 2022-06-18 04:26:40.000001 UTC

SAM\Domains\Account\Users\000003E9

Last written 2025-01-02 08:41:00.000000 UTC · FILETIME 133802808600000000 · 5 value(s) · cell 0x680

Name	Type	Size	Interpreted
F	REG_BINARY	80	**Last logon:** 2025-03-14 09:12:00 UTC **Password last set:** 2025-01-02 08:41:00 UTC **Account expires:** not a date: stored as 0x7FFFFFFFFFFFFFFF **Last failed logon:** Never (0) **RID:** 1001 (0x3e9) **Account flags:** 0x0210: Normal user account, Password does not expire **Failed logon count:** 0 **Logon count:** 37
V	REG_BINARY	260	**User name:** jdoe **Full name:** Jordan Doe **Profile path:** C:\Users\jdoe **LM hash entry:** 0 bytes (not decoded here) **NT hash entry:** 4 bytes (not decoded here)
InternetUserName	REG_BINARY	44	6a 00 6f 00 72 00 64 00 61 00 6e 00 2e 00 64 00 6f 00 65 00 40 00 65 00 78 00 61 00 6d 00 70 00 6c 00 65 00 2e 00 63 00 6f 00 6d 00
ForcePasswordReset	REG_BINARY	4	00 00 00 00
UserTile	REG_BINARY	6982	**Picture:** BMP 48x48, 6966 bytes at offset 16

SAM\Domains\Builtin\Aliases\00000220

Last written 2025-03-14 09:30:00.000000 UTC · FILETIME 133864182000000000 · 1 value(s) · cell 0x29f0

Name	Type	Size	Interpreted
C	REG_BINARY	272	**Group name:** Administrators **Description:** Administrators have complete and unrestricted access to the computer **Members:** S-1-5-21-2111111111-3222222222-1333333333-500 S-1-5-21-2111111111-3222222222-1333333333-1001

Key times are last-written times from the hive, in UTC. Interpreted fields come only from SAM layouts with a published source (listed in regfprobe_interpret.py) and from pictures whose file header checks out; every other value is shown as regfprobe decodes it. Raw bytes are shown as stored.

A report on one account and the Administrators group, without the optional columns. It follows the reader's light or dark setting.

10. The Interpreted column

Many important registry values are binary structures that regfprobe's Values tab can only show as hex. The Interpreted column in reports decodes the ones below, using only layouts with a published source (the sources are listed at the top of `regfprobe_interpret.py`). Nothing is guessed from a value's name or

from what its bytes look like. Any other value keeps regfprobe's own decoding (for binary data, its hex), shown here when the regfprobe data column is left out and as a dash when that column is included.

Where	Value	Decoded into
SAM `Domains\Account\Users\<RID>`	`F`	Last logon, password last set, account expiry, last failed logon, RID, account flags (disabled, password not required, password never expires...), failed and total logon counts
SAM `Domains\Account\Users\<RID>`	`V`	User name, full name, comment, user comment, home directory and drive, logon script, profile path, allowed workstations; the size of the LM and NT hash entries (never the hashes themselves)
SAM `Users\Names\<name>`	(default)	The account's RID. Windows stores it as the value's *type*, which is why the Values tab shows "unknown type 0x3e9".
SAM `Builtin\Aliases\<RID>` and `Account\Aliases\<RID>`	`C`	Group name, description and member SIDs
SAM `Domains\Account\Users\<RID>`	`UserTile`	The account picture, found after the value's header and shown in the report
Anywhere, binary values	Data that starts with a PNG or BMP header that checks out, or with a JPEG's first three bytes	The picture itself

> **Read SAM logon data with care.** The last logon, logon counts and other F fields are shown as stored. Which sign-ins update them is not covered here, so "Never" or 0 does not prove the account was never used; check it against the event logs.

11. Key profiles

A profile is a saved list of keys you look at on most cases. Save it once, then pull all of those keys from any new image in one step, ready to report on.

Saving a profile

1. Add your usual keys to the report list (section 9), from any hives.
2. **Report → Save Selections as Profile...** and choose a name, e.g. `System Information.json`.

Using a profile on a new case

1. **Report → Load Profile from Disk Image...**
2. Choose the profile, then the E01 or raw image.
3. regfprobe searches the image once, reads each hive the profile needs (with its logs), and adds every key to the report list.
4. A message lists anything the profile asked for that this image does not have; the Keys in Report window then opens so you can review and export.

Loading is additive: load several profiles in a row, or add more keys by hand before exporting.

Profiles fit any machine

A profile records *where* a key lives, not which case it came from. Parts that change from machine to machine are saved as patterns and filled in when the profile is loaded:

You picked	The profile saves	On a new image, that means
`Users/jdoe/NTUSER.DAT`	`Users/*/NTUSER.DAT`	That hive for **every** user (UsrClass.dat likewise). The Default profile's hive is checked too.
SAM `Users\000003E9`	SAM `Users\*`	Every account's key
SAM `Users\Names\jdoe`	SAM `Users\Names\*`	Every account name
SYSTEM `ControlSet001\...`	SYSTEM `CurrentControlSet\...`	The control set `Select\Current` names on that machine. This happens only when the set you picked was the current one; any other control set is saved as picked.

Loading uses each key exactly as the profile file names it; nothing is rewritten on load.

The profile file

A profile is a small, readable JSON file. You can share it with colleagues or edit it by hand; `*` stands for any one folder (in hive paths) or any one subkey (in key paths).

```
{
  "regfprobe_profile": 1,
  "name": "System Information",
  "entries": [
    {"hive": "Windows/System32/config/SOFTWARE", "key": "Microsoft\\Windows
NT\\CurrentVersion", "subkeys": false},
    {"hive": "Windows/System32/config/SYSTEM", "key":
"CurrentControlSet\\Control\\TimeZoneInformation", "subkeys": false},
    {"hive": "Windows/System32/config/SAM", "key":
"SAM\\Domains\\Account\\Users\\*", "subkeys": false},
    {"hive": "Users/*/NTUSER.DAT", "key":
"Software\\Microsoft\\Windows\\CurrentVersion\\Explorer\\RecentDocs", "subkeys":
false}
  ]
}
```

A hive opened as a loose file is saved under its usual place in an image when its name is a standard one (SAM, SYSTEM, SOFTWARE, SECURITY, DEFAULT, NTUSER.DAT, UsrClass.dat, Amcache.hve); others are left out, and saving says so.

> Each hive a report uses stays in memory until regfprobe closes. That is small for SAM, SYSTEM and user hives; a large SOFTWARE hive can take a few hundred MB.

12. Dark mode

The **Light** and **Dark** switch at the top right of the window, or **View** → **Dark Mode**, switches the whole window and its dialogs (and the title bars on Windows), and the choice is remembered for next time. Until a choice is made the window follows the system's setting where it can read it (macOS, Windows, and GNOME's setting on Linux), and is otherwise light. Every screenshot in this manual was taken in dark mode. The menu bar stays light because the operating system draws it. HTML reports follow the browser's own light/dark setting; this manual does not.

13. Keyboard shortcuts

On macOS, use `Cmd` where this table says `Ctrl`.

Keys	Action
`Ctrl` + `O`	Open a hive file
`Ctrl` + `Shift` + `O`	Open a hive inside a disk image
`Ctrl` + `L`	Go to a key by path
`Ctrl` + `F`	Search
`Ctrl` + `C`	Copy the selected value's data
`Ctrl` + `Shift` + `C`	Copy the key path
`Ctrl` + `E`	Export the selected key (JSON, CSV, .reg)
`Ctrl` + `R`	Add the selected key to the report
`Ctrl` + `1` … `Ctrl` + `4`	Values, Security, Search, Timeline tabs
`F1`	Open this manual
`Esc`	Cancel a running task
Arrows, `Enter`, `Tab`	Move in the tree, open or close a key, move between panes
`Shift` + `F10`, or the Menu key on Windows and Linux	The right-click menu of the selected row

14. Command line

`regfprobe.py` also works without the window, which suits scripts and batch work. Each command has its own `--help`.

Command	Does
`info`	Base block, recovery state and counts
`ls`	Subkeys and values of a key
`cat`	Decoded data and hex of a key's values
`find`	Search key names, value names and data
`timeline`	Every key by last-written time, as CSV
`export`	Keys and values as JSON, CSV or .reg
`logs`	What the transaction logs hold and what replay changes
`hives`	List files named like hives inside a disk image

```
python regfprobe.py info SYSTEM
python regfprobe.py ls SOFTWARE "Microsoft\Windows NT\CurrentVersion"
python regfprobe.py export SYSTEM "ControlSet001\Services" --format csv --out
services.csv
python regfprobe.py hives --image "D:\Cases\CASE-001.E01"
python regfprobe.py logs SYSTEM
```

15. Tips and cautions

- **Read-only.** regfprobe never changes evidence. Transaction-log replay happens in memory only.
- **Times are UTC.** Convert them yourself using the machine's time zone (SYSTEM `CurrentControlSet\Control\TimeZoneInformation`), which is a good key to keep in a profile.
- **Dirty hives.** `regfprobe.py logs` shows what the transaction logs hold and what replay changes. When the logs are there, opening the hive with replay on and off shows the same difference in the window.
- **Interpreted fields are an aid.** Reports keep the raw bytes available (optional column) so any decoding can be checked.
- **Not found is not absence.** When a profile reports a key missing, the key may simply be named differently on that Windows version.

Screenshots use a made-up demo hive (`docs/demo/SAM`, fictional accounts). To retake them after a change: `python docs/make_demo_hive.py` then `python docs/make_screenshots.py`. To rebuild the PDF: `python docs/make_pdf.py`.